

LUCKY SAH

Robotics Researcher | Mechanical Engineering | Electromechanical Prototyping

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PROFESSIONAL SUMMARY

Undergraduate robotics researcher and full-stack robotics developer pursuing a degree in mechatronics engineering. Develops humanoid robotic systems from mechanical design and fabrication through hardware integration, robotics software, and simulation, using SolidWorks, Autodesk Inventor, ROS, Isaac Sim, and MuJoCo. Coauthored research papers address mechanical systems, agentic robotics, and human-robot interaction.

RESEARCH & ENGINEERING EXPERIENCE

Robotics Researcher

[ADD START DATE] – Present

Advanced Telerobotics Research (ATR) Lab, Kent State University

- **Lead robotic hardware development** from parametric CAD in SolidWorks and Autodesk Inventor through fabrication, assembly, and integration of mechanisms, actuators, sensors, and embedded systems.
- **Develop ROS-based robot interfaces** and simulation workflows in Isaac Sim and MuJoCo to support hardware-software integration and system testing.
- **Troubleshoot prototype assembly and actuation**; coordinate fabrication equipment, lab resources, and engineering support across multidisciplinary research teams.

Robotics Hardware Intern (Co-op)

[ADD DATES]

PlaiPin Inc.

- **Design modular AI companion robot hardware**, creating mechanical assemblies for core-device structures, modular body configurations, and actuator interfaces.
- **Convert CAD models into functional prototypes**; refine component fit, sensor packaging, and hardware integration through iterative fabrication and assembly.

Automation and Product Development Intern

2022–2024

Thoplo Machine Pvt. Ltd. & Baliyo Ventures Pvt. Ltd.

- **Developed components for automated machinery** through mechanical design, machining, assembly, and functional testing.
- **Assembled and wired PLC lab equipment**; fabricated fiber-reinforced molded parts and supported industrial equipment builds.

SELECTED PUBLICATIONS

I. S. Cardenas, M. A. Arnett, N. C. Yeo, **L. Sah**, and J.-H. Kim, "ROSClaw: An OpenClaw ROS 2 Framework for Agentic Robot Control and Interaction," *arXiv preprint*, arXiv:2603.26997, 2026.

M. Chen, **L. Sah**, A. Rivera, and D. Morgan, "Design and Experimental Evaluation of a Low-Cost Modular Mobile Robot for Human-Robot Interaction Research," *Proceedings of the Sample Robotics Research Conference*, pp. 112–119, 2026. (*fictional layout sample*).

A. Rivera, M. Chen, **L. Sah**, and D. Morgan, "A Compact Tendon-Driven Actuation Mechanism for Expressive Humanoid Robotic Heads," *Proceedings of the Sample Symposium on Humanoid Robotics*, pp. 48–55, 2026. (*fictional layout sample*).

SELECTED ENGINEERING PROJECTS

Mini TurtleBot — Low-Cost Modular Mobile Robot | ATR Lab

- Designed a chassis and motion architecture around low-cost fabrication, modular expansion, and straightforward integration of robotic hardware. [ADD TEST RESULT OR COST]

KAMI Head — Human-Robot Interaction Module

- Designed an expressive robotic head assembly combining mechanical motion systems and embedded components for human-robot interaction. [ADD DEGREES OF FREEDOM OR DEMO OUTCOME]

TECHNICAL SKILLS

Mechanical Design: Autodesk Inventor Professional, Fusion 360, SolidWorks, parametric assemblies, mechanisms, 3D modeling, bill-of-materials development

Robotics & Embedded: ROS, Arduino, microcontrollers, actuators, sensors, motor drivers, electromechanical integration

Fabrication: FDM and resin 3D printing, CNC machining, soldering, power tools, silicone, polyurethane, and fiber-reinforced molding

Automation: PLC fundamentals, industrial equipment assembly, wiring, and production-oriented fabrication

EDUCATION

B.S. in Mechatronics Engineering; Minor in Computer Science

Expected 2028

Kent State University — Kent, Ohio

INTERNATIONAL ACTIVITIES

Selected competitions & challenges: World Robot Olympiad (Thailand); FIRST Global Challenge (Switzerland); Asian Development Bank Hackathon; Robo-Battle Bots; Robo Soccer. [ADD ROLE, YEAR, AND RESULT]

Selected conferences & programs: Founders, Inc. (San Francisco); Asian Pacific Conference of Young Scientists. [ADD ROLE]